

MIAE 221: Materials Science for Engineers

Chapter 5: Diffusion in Solids (Expanded & Condensed Guide)

Concordia University · Gina Cody School of Engineering

Textbook:

Materials Science and Engineering: An Introduction

(10th Ed., Callister & Rethwisch)

Instructor: Dr. Mamoun Medraj, P.Eng | **Curriculum Alignment:** Concordia University

(Syllabus Week 5 · Core Midterm Topic)

1. 🌀 30-Second Core Intuition & Plain-English Concept

Diffusion is the mass transport of atoms through a solid by random atomic jumping driven by thermal vibrations. Atoms jump into neighboring vacancies (**vacancy diffusion**) or squeeze through interstices (**interstitial diffusion**).

Because small solute atoms (like C, H, N) don't need to wait for a vacancy to open up, **interstitial diffusion is thousands of times faster** than substitutional diffusion.

Real-World Analogy

: Squeezing through a packed concert crowd. If you are a child (small interstitial carbon atom), you can weave between people's legs quickly. If you are a large adult (substitutional atom), you can only take a step forward when someone in front of you leaves their spot (vacancy diffusion).

2. ⚙️ High-Yield Mathematical Engine & Essential Laws

1. Fick's First Law (Steady-State Diffusion)

$$J = -D \frac{dC}{dx}$$

- J : Diffusion flux ($\text{extkg}/(\text{extm}^2 \cdot \text{s})$ or $\text{extatoms}/(\text{extm}^2 \cdot \text{s})$).
- D : Diffusion coefficient (extm^2/s).
- $\frac{dC}{dx}$: Concentration gradient (extkg/m^4). The negative sign indicates diffusion flows down the concentration gradient (from high to low concentration).

2. Fick's Second Law (Non-Steady-State Diffusion)

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$$

Standard Solution for Semi-Infinite Solid with Constant Surface Concentration C_s

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

- C_0 : Uniform initial bulk concentration.
- C_s : Constant surface concentration.
- C_x : Concentration at depth x after elapsed time t .
- $\operatorname{erf}(z)$: Gaussian error function.
- **Golden Shortcut Rule:** When concentration ratio $\frac{C_x - C_0}{C_s - C_0}$ is held constant:

$$\frac{x^2}{Dt} = \text{constant} \implies \frac{x_1^2}{D_1 t_1} = \frac{x_2^2}{D_2 t_2}$$

3. Temperature Dependence of Diffusion (Arrhenius)

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right) \iff \ln D = \ln D_0 - \frac{Q_d}{R} \left(\frac{1}{T}\right)$$

- Q_d : Activation energy for diffusion (*extJ/mol*).
- R : Universal gas constant (8.314 J/mol · K).

3. Textbook Reference Diagram & Visual Pedagogical Analysis

(b) Note the kind of diffusion for which each of these equations is normally applied. diffusion constants.

5.1 INTRODUCTION

diffusion

WileyPLUS

Tutorial Video:
What Is Diffusion?

Many reactions and processes that are important in the treatment of materials rely on the transfer of mass either within a specific solid (ordinarily on a microscopic level) or from a liquid, a gas, or another solid phase. This is necessarily accomplished by **diffusion**, the phenomenon of material transport by atomic motion. This chapter discusses the atomic mechanisms by which diffusion occurs, the mathematics of diffusion, and the influence of temperature and diffusing species on the rate of diffusion.

The phenomenon of diffusion may be demonstrated with the use of a *diffusion couple*, which is formed by joining bars of two different metals together so that there is intimate contact between the two faces; this is illustrated for copper and nickel in Figure 5.1a, which includes schematic representations of atom positions and composition across the interface. This couple is heated for an extended period at an elevated temperature (but below the melting temperature of both metals) and cooled to room temperature. Chemical analysis reveals a condition similar to that represented in Figure 5.1b—namely, pure copper and nickel at the two extremities of the couple, separated by an alloyed region. Concentrations of both metals vary with position as shown in Figure 5.1b (bottom). This result indicates that copper atoms have migrated or diffused into the nickel, and that nickel has diffused into copper. The process by which atoms of one metal diffuse into another is termed **interdiffusion**, or **impurity diffusion**.

interdiffusion

Figure 5.1: Copper-Nickel diffusion couple: (a) Schematic atom positions before heating, and (b) Concentration profile across the interface after elevated-temperature diffusion.

where C_x represents the concentration at depth x after time t . The expression $\text{erf}(x/2\sqrt{Dt})$ is the Gaussian error function,⁴ values of which are given in mathematical tables for various $x/2\sqrt{Dt}$ values; a partial listing is given in Table 5.1. The concentration parameters that appear in Equation 5.5 are noted in Figure 5.5, a concentration profile taken at a specific time. Equation 5.5 thus demonstrates the relationship between concentration, position, and time—namely, that C_x , being a function of the dimensionless parameter $x/\sqrt{Dt}$, may be determined at any time and position if the parameters C_0 , C_s , and D are known.

Suppose that it is desired to achieve some specific concentration of solute, C_1 , in an alloy; the left-hand side of Equation 5.5 now becomes

$$\frac{C_1 - C_0}{C_s - C_0} = \text{constant}$$

considered to be semi-infinite if none of the diffusing atoms reaches the bar end during the time t takes place. A bar of length l is considered to be semi-infinite when $l > 10\sqrt{Dt}$. The error function is defined by

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-y^2} dy$$

where y has been replaced by the variable z .

Figure 5.5: Concentration profile C_x vs. depth x into a solid during gas carburizing at a specific time t .

In-Depth Visual Breakdown:

- **The Diffusion Couple (Figure 5.1):** Illustrates how an initially sharp chemical step-function (100 on left, 100 on right) gradually smooths out into an S-shaped continuous concentration curve as Cu atoms diffuse right and Ni atoms diffuse left.
- **Carburizing Profile (Figure 5.5):** Shows steel surface hardening. Carbon gas at surface maintains constant C_s . As diffusion proceeds, the carbon profile pushes deeper into the interior. The depth x at which a target hardness/carbon level is reached scales directly with $\sqrt{Dt}$.

4. Teacher Notes Cross-Reference & Concordia Exam Traps

- **Concordia Exam Focus:**
 - Gas carburizing calculations for steel gear teeth: Determining time required to achieve a specified carbon concentration at a given depth.
 - **Linear Interpolation of Error Function Table:** Concordia exams provide an abbreviated table of $\text{erf}(z)$ values; students must interpolate precisely:

$$z = z_1 + \frac{\text{erf}(z) - \text{erf}(z_1)}{\text{erf}(z_2) - \text{erf}(z_1)}(z_2 - z_1)$$

- **Concordia Exam Traps:**
 - **Gas Constant Units:** Using k_B instead of R . If Q_d is given in kJ/mol, use $R = 8.314 \text{ J/mol} \cdot \text{K}$ (multiply kJ by 1000!). If Q_d is in eV/atom, use $k_B = 8.62 \times 10^{-5} \text{ eV/K}$.

5. Master Exam Problem & Step-by-Step Solution Framework

Problem:

A gear made of 0.20 wt carbon steel is case-hardened in a gas atmosphere maintaining 1.20 wt carbon at the surface at 950°C ($D = 1.6 \times 10^{-11} \text{ m}^2/\text{s}$). How long (in hours) will it take to achieve a carbon concentration of 0.60 wt at a depth of 0.5 mm below the surface?

$$(\text{Given: } \text{erf}(0.60) = 0.6039, \text{erf}(0.65) = 0.6420)$$

- **Step 1: Set Up Fick's Second Law Concentration Ratio**

$$\frac{C_x - C_0}{C_s - C_0} = \frac{0.60 - 0.20}{1.20 - 0.20} = \frac{0.40}{1.00} = 0.40$$

- **Step 2: Solve for Error Function Value**

$$1 - \text{erf}(z) = 0.40 \implies \text{erf}(z) = 0.60$$

- **Step 3: Interpolate to Find Argument z**

$$z = 0.60 + \frac{0.6000 - 0.6039}{0.6420 - 0.6039}(0.65 - 0.60) \approx 0.595$$

- **Step 4: Relate z to Physical Diffusion Parameters**

$$z = \frac{x}{2\sqrt{Dt}} \implies 0.595 = \frac{0.5 \times 10^{-3} \text{ m}}{2\sqrt{(1.6 \times 10^{-11} \text{ m}^2/\text{s})t}}$$
$$\sqrt{t} = \frac{0.5 \times 10^{-3}}{2(0.595)\sqrt{1.6 \times 10^{-11}}} = \frac{5 \times 10^{-4}}{4.757 \times 10^{-6}} = 105.1 \text{ s}^{1/2}$$
$$t = (105.1)^2 = 11,048 \text{ seconds} = \frac{11,048}{3600} = 3.07 \text{ hours}$$
